

Key for Log Files Headers

Non-hypersonic logs

LAMMPS log column	Excel column
Step	Number of Timesteps
TotEng	Total Energy (eV)
KinEng	Total Kinetic Energy (eV)
PotEng	Total Potential Energy (eV)
Density	Density (g/cm ³)
Temp	Global Temperature (K)
c_obsTemp	Temperature of Physisorption Region (K)
c_obs_TiTemp	Temperature of the Titanium Surface Atoms Only (K)
c_T_middle	Temperature of Middle Titanium Layer (K)
c_obs_O2Temp	Temperature of the Oxygen Molecules Only (K)
Press	Total System Pressure (bar)
v_O2pressure	Pressure of Oxygen Molecules Only (bar)
v_n_obs	Number of Oxygen Atoms in the Physisorption Region

Hypersonic logs

LAMMPS log column	Excel column
Step	Number of Timesteps
TotEng	Total Energy (eV)
KinEng	Total Kinetic Energy (eV)
PotEng	Total Potential Energy (eV)
Density	Density (g/cm ³)
Temp	Global Temperature (K)
Press	Total System Pressure (bar)
c_obsTemp	Temperature of Physisorption Region (K)
c_obs_TiTemp	Temperature of Titanium Surface Atoms Only (K)
v_n_obs	Number of Oxygen Atoms in Physisorption Region
c_T_O2	Temperature of Oxygen Molecules Only (K)
c_T_O2_nonxh	Temperature of Non-Hypersonic Oxygen Molecules (K)
v_nout	Number of Oxygen Atoms in Outlet Region